

Reference	Quantity	Value Case Code – Inches	Description	Specification	Mouser Part Number	Remarks / Alternate Parts
Resistors						
R28, R32, R37	3	0R1 SMD 1206	Current Sense Resistors	0.5% 0.5W	588-KDV12DR100ET	652-CR0603-F/1R00ELF
R1, R3	2	1R SMD 0603	Thick Film Resistors	1% 0.1W	603-RC0603FR-071RL	
R18, R24, R25,	3	100R SMD 0603	Thick Film Resistors	1% 0.1W	603-RC0603FR-13100RL	
R14, R15, R16, R17, R20	5	1k SMD 0603	Thick Film Resistors	1% 0.1W	603-RC0603FR-071KL	
R2, R4, R10, R38	4	10k SMD 0603	Thick Film Resistors	1% 0.1W	652-CR0603FX-1002ELF	
R5, R6,	2	1K2 SMD 0603	Thick Film Resistors	1% 0.1W	603-RC0603FR-101K2L	
R21, R29, R30, R31, R34, R36	6	1k68 SMD 0603	Thick Film Resistors	1% 0.1W	603-RC0603FR-071K69L	
R12, R13, R19,	3	2k SMD 0603	Thick Film Resistors	1% 0.1W	603-RC0603FR-072KL	
R22,	1	22K SMD 0603	Thick Film Resistors	1% 0.1W	603-RC0603FR-1322KL	
R7, R23	2	240R SMD 0603	Thick Film Resistors	1% 0.1W	603-RC0603FR-13240RL	
R9,	1	274R SMD 0603	Thick Film Resistors	1% 0.1W	754-RR0816P-2740D43A	667-ERA-3AHD330V
R8, R11,	2	33R SMD 0603	Thick Film Resistors	0.5% 0.1W	603-RC0603FR-10330RL	
R26, R27, R33	3	330R SMD 0603	Thick Film Resistors	1% 0.1W	603-RC0603FR-10470RL	
R35	1	470R SMD 0603	Thick Film Resistors	1% 0.1W		
Capacitors						
C32, C33,	2	100pF SMD/SMT 0603	Multilayer Ceramic Capacitors MLCC	10V 5%	710-885012006008	81-GRM155R71H103KA8J
C9, C17, C19, C20,	4	1nF SMD/SMT 0603	Multilayer Ceramic Capacitors MLCC	25V 5% X7R	603-CC603JRX7R8BB102	
C37, C40, C41, C44, C47,	5	10nF SMD/SMT 0603	Multilayer Ceramic Capacitors MLCC	50V 10%	603-CS0603KRX79BB103	
C51,	1	10nF SMD/SMT 0402	Multilayer Ceramic Capacitors MLCC	50V 10% X7R		
C1, C2, C8, C10, C11, C14, C15, C16, C18, C21, C22, C24, C25, C28, C30, C31, C34, C36, C38, C39, C42, C46, C48, C54, C55, C57, C58, C63,	28	100n SMD/SMT 0603	Multilayer Ceramic Capacitors MLCC	10V 5% X7R	80-C0603C104J4RAC	
C3, C6, C7, C23, C26, C27, C43, C45, C50,	9	1uF SMD/SMT 0603	Multilayer Ceramic Capacitors MLCC	25V +/-10% X5R	187-CL10A105KA8NNNC	581-06033C221K4T2A
C4, C5, C62,	3	1uF SMD/SMT 1206	Tantalum Capacitors	16V 10% Tantalum	80-T491A105K016	
C29, C35,	2	10uF SMD/SMT 0603	Multilayer Ceramic Capacitors MLCC	25V 10% X5R	81-GRM188R61E106KA3D	
C53, C56, C59, C60, C61,	5	10uF SMD/SMT 0805	Multilayer Ceramic Capacitors MLCC	25V +/-10% X5R	963-TMK212BBJ106KG-T	
C13,	1	18nF SMD/SMT 0603	Multilayer Ceramic Capacitors MLCC	25V 10% X7R	603-CC603KRX7R8BB183	
C12,	1	22pF SMD/SMT 0603	Multilayer Ceramic Capacitors MLCC	16V 5%	710-885012006019	
C52,	1	220pF SMD/SMT 0603	Multilayer Ceramic Capacitors MLCC	25V 10% X7R	603-CC603KRX7R8BB221	
C49,	1	2u2F SMD/SMT 0603	Multilayer Ceramic Capacitors MLCC	16V 10% X7S	810-C168X7S1C225K8AC	
Inductors						
L1	1	120nH SMD 0402	Fixed Inductors	1600mOhms +/-5% 150mA	710-7447840212G	279-BMC-Q1JY1000H
L2, L3, L4, L5, L6	5	1k500mA SMD 0603	Ferrite Beads	1000Ohm 500mA	704-MFBW1V1608-102-R	
Diodes						
D2, D4, D6, D9, D10, D12,	6	green SMD 0603	Standard LEDs	Water Clear 570nm 40mcd 120deg	604-APT1608CGCK	D3, D13 follow placement instructions
D3, D13	2	1N4148 SOD-323F	Diodes – General Purpose	Switching Small Signal Diode	512-1N4148WS	

D7,	1	BAV99 SOT-23-3	Diodes – General Purpose	Power, Switching 75V 350mW	757-BAV99LM	
D8,	1	3V9 SOT-23-3	Zener Diode	3.9V 225mW	863-MMBZ5228BLT1G	
D1, D5, D11	3	5V1 SOT-23-3	Zener Diode	5.1V 225mW	863-MMBZ5231BLT3G	D11 Do not Populate
Transistors						
Q3,	1	BSS138L SOT-23-3	N-channel MOSFET	50V 0.20A	512-BSS138L	863-BSS138LT3G
Q2,	1	IRLML6246 SOT-23-3	MOSFET	20V 4.1A	942-IRLML6246TRPBF	
Q1,	1	IRF7807Z SOIC-8	Power MOSFET	30V 11A	942-IRF7807ZTRPBF	942-IRF7855TRPBF
DC/DC Converters						
U1,	1	RB-0509D SIP-7	Isolated DC/DC Converter	1W 5Vin +/-9Vout	919-RB-0509D	
U2,	1	REE-0505S SIP-7	Isolated DC/DC Converter	1W 5Vin 5Vout	919-REE-0505S	
Regulators						
U5	1	LD59015 SOT-323-5	Linear Voltage Regulators	3.3V 150 mA	511-LD59015C33R	
U12	1	LD1117S33 SOT-223-3	LDO Voltage Regulators	3.3V 0.8A Positive	511-LD1117S33	
U3	1	LM317L SOIC-8	Linear Voltage Regulators	100mA 35V Adjustable Positive	863-LM317LBDR2G	
U4	1	LM337 SOIC-8	Linear Voltage Regulators	Adjustable Negative	926-LM337LMX/NOPB	
U6	1	LP3985IM5-3.0 SOT-23-5	LDO Voltage Regulators	3.0V 150ma Low Noise	926-3985IM5-3.0/NOPB	
U8	1	TPS7A7100 VQFN-20	LDO Voltage Regulators	1A,Sgl Out,Very Low Inp,Config Fixed	595-TPS7A7100RGWR	
ICs						
U10, U11, U13	3	INA225 MSOP-8	Current Sense Amplifiers	36V Bidir 0-Drift Ser Crnt-Shunt Mon	595-INA225AIDGKT	595-INA225AIDGKR
U7	1	AD8331 QSOP20	Special Purpose Amplifiers	SGL VGA w/ Ultra low Noise	584-AD8331ARQZ	
U9	1	AD9215 TSSOP28	Analog to Digital Converters	ADC 10-Bit 105 MSPS 3V	584-AD9215BRUZ-105	
Connectors						
J7, J9, J15, J18, J19, J20, J21, J24, J26, J27, J28, JP2, JP3	1	Pin Header Through Hole 1x36	Headers & Wire Housings	36Pin, 1Row, Breakaway header, Mating Post Length: 5.84 mm, gold plated	649-68001-236HLF	break strip into matching headers, J7 only 4 pin, total 18x1
J3, J4, J6, J8	1	Pin Header Through Hole 2x36	Headers & Wire Housings	72Pin, 2Row, Breakaway header, Mating Post Length: 5.84 mm, gold plated	649-67997-272HLF	break strip into matching headers, J6 covers 4 pins of J7, total 26x2
J16, J22	2	Test Points Through Hole	Circuit Board Hardware	PCB test point, black	534-5281	
J17	1	Barrel Jack Through Hole	DC Power Connectors Power Jacks	Inside Contact Diameter: 2 mm, Outside Contact Diameter: 6.5 mm	490-PJ-102AH	
J2, J10, J13, J14	4	SMA Edge Connector	RF Connectors / Coaxial Connectors	SMA 50 Ohm End Launch Jack Receptacle - Tab Contact, gold plated Board Thickness 0.062	142-0701-851	
J23	1	SMA Through Hole	RF Connectors / Coaxial Connectors	SMA 50 Ohm Female PCB Mnt, Extended, Gold plated	712-CONSMA008-G	
J11, J12	2	Shrouded Header Through Hole 2x3	Headers & Wire Housings	6Pin, 2 Rows, right angle, shrouded, gold plated	649-75867-131LF	
J1	1	Shrouded Header Through Hole 2x10	Headers & Wire Housings	20Pin, 2 Rows, right angle, shrouded, gold plated	517-30320-5002	649-75867-304LF
J5	1	Pin Socket Through Hole 2x3	Headers & Wire Housings	6Pin, 2 Rows, straight, gold plated	517-929852-01-03-RA	
J25	1	Pin Header, Locking Through Hole 1x2	Headers & Wire Housings	Molex KK-254, vertical, gold plated	538-22-11-2022	
H1	1	Heat Sink TO-252	Heat Sink	Extruded Fin, Length: 8 mm, Width: 23 mm, Height: 10 mm	532-573100D00010G	

Cables Control to DUT					
Target Cable	2	IDC Socket IDC 2x10	Headers & Wire Housings	20Pin, 2 Rows, straight, IDC cable mount, with index, gold plated	649-71600-020LF
Target Cable	1	Ribbon Cable 20 Conductors, 1ft	Flat Cables	20 Conductors, 28AWG	517-3365/20FT
SMA Cable	1	SMA Cable RG-316, 2ft	RF Cable Assemblies	SMA to SMA cable RG-316, 2ft, gold plated	530-415-0029-024

Part References in red indicate a discrepancy from the published schematic. Please follow this BOM as it is more up to date.

Last Modification	4 Nov 25
Board version	Rev 2